Free energy and kinetic theory from the S-matrix

Based on *Free energy from forward scattering in 1+1d* (2024), arXiv:2408.00729

Daniel Schubring

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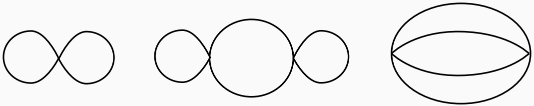
Free energy via thermal field theory

- Free energy is a basic QFT observable, depending on temperature (optionally chemical potentials).
- Standard method: compactify time to length β (Euclidean), compute via vacuum bubble diagrams.
- In $1+1$d: finite-temperature free energy $\sim$ finite-volume Casimir energy.



Free energy via TBA

- In $1+1$d: free energy (Casimir energy) can also be computed using the Thermodynamic Bethe Ansatz **(TBA)**.
- For one-particle spectra (e.g. sinh–Gordon), the TBA gives:

$$f = -T \int \frac{dp}{2\pi} \log\left(1 + e^{-\epsilon(p)/T}\right), \quad \epsilon(p) = E(p) - \mu - T \log\left(1 + e^{-\epsilon/T}\right) \star K$$

  where $K(p) = \frac{1}{2\pi i} \frac{d}{dp} \log S(p)$.
- At zeroth order, K reduces to a delta function $\Rightarrow$ reproduces the free boson gas.
- Even allows for analytic continuation to excited states (e.g. Dorey–Tateo).
- But applicability is limited to **integrable theories**.

Another approach

- Third method: compute density of states from the S-matrix

$$Z = \int dE\, e^{-\beta E}\, \rho(E), \qquad \rho(E) = \rho^{(0)}(E) + \tfrac{1}{2\pi i}\tfrac{d}{dE} \mathrm{Tr}\, \log \hat{S}(E)$$

- $\hat{S}(E)$: S-matrix operator (details later). – Resembles TBA kernel, but applies also to **non-integrable theories**.
- Historical development:
  - Dashen–Ma–Bernstein (1969): general formula (DMB)
  - Norton (1981–86): DMB from thermal field theory
  - Jeon & Ellis (1998): DMB for the self-energy
  - DS (2024), Baratella–Miró–Gendy (2024)
- Approach remained niche. Multiparticle states were not well-understood.

The confining string on the lattice

- Both DMB papers in 2024 inspired by the effective string theory program
- SU(N) gauge theory has confining flux tubes in 3+1d and 2+1d
- Athenodorou, Bringoltz, Teper (2011): Energy levels from Wilson line correlators
- The Wilson lines wrap a periodic dimension of length L
- Can a theory of the flux tube calculate the energy levels as a function of L?

Effective string theory

- As $N \to \infty$ the flux tube dynamics decouples from the bulk
- We can form an EFT from the embedding coordinate fields X^μ
- The lowest order term is Nambu-Goto, just as for fundamental strings
- In $D = 3$, static gauge

$$\mathcal{L} = \frac{1}{l^2}\sqrt{1 + l^2 (\partial X)^2} = \frac{1}{l^2} + \frac{1}{2}(\partial X)^2 - \frac{l^2}{8}(\partial X)^4 + \frac{l^4}{16}(\partial X)^6 + \dots$$

- Higher order operators matter at much higher order in l^2/L^2

GGRT model

- Exact results are known in an integrable case
- Goddard–Goldstone–Rebbi–Thorn (GGRT): Light cone quantization of NG

$$E_N(L) = \frac{L}{l^2}\sqrt{1 + 8\pi\frac{l^2}{L^2}\left(N - \frac{D-2}{24}\right)}$$

- Dubovsky, Flauger, Gorbenko (2012)
  - Exact S-matrix $S = e^{il^2 s/4}$
  - E_N derived from TBA
- However flux tubes in Yang-Mills are non-integrable!
- Basic problem: Can we go from a non-integrable S-matrix to $E_N(L)$?

Lüscher's many formulas

- "Lüscher term" $\to$ Casimir energy of non-interacting massless bosons.
- M. Lüscher, *Volume dependence of the energy spectrum in massive QFTs I & II*.
  - Part I: finite-volume shift of **single-particle** energies.
  - Part II: finite-volume **two-particle** energies via the S-matrix.
- In $1+1$d: two-particle Lüscher formula $=$ asymptotic Bethe ansatz:

$$2\pi i n = i k_n L + 2\log S(k_n)$$

- Phase around circle $+$ scattering phase must equal $2\pi n$.
- Valid up to exponentially small corrections $\sim e^{-mL}$.

Beth–Uhlenbeck formula

$$\frac{n}{2} = \frac{L}{4\pi} k_n + \frac{1}{2\pi i} \log S(k_n),$$

- Beth & Uhlenbeck (1937)

$$\begin{aligned} \rho_{(2)}(E) &= \left(\frac{dE}{dk}\right)^{-1} \frac{1}{k_{n+2} - k_n} \\ &= \frac{L}{4\pi} \left(\frac{dE}{dk}\right)^{-1} + \frac{1}{2\pi i} \frac{d}{dE} \log S(E) \end{aligned}$$

- Limitation: applies only to 2-body scattering.
- DMB (1969): generalized to multi-particle and general QFTs.

Old-fashioned perturbation theory

- Start from Lippmann–Schwinger:

$$\Psi^{\pm} = \Phi + \frac{1}{E - H_0 \pm i\epsilon} V \Psi^{\pm}.$$

- Use to define unitary operators $\Psi^{\pm} = \Omega^{\pm}\Phi$
- Write in terms of resolvent (propagator):

$$\Omega^{+} = (1 - G_0 V)^{-1}, \qquad G_0(E) \equiv \frac{1}{E - H_0 + i\epsilon}$$

- This lets us define scattering operator explicitly

$$\hat{S} = \left(\Omega^{-}\right)^{\dagger} \Omega^{+} = \left(1 - G_0^{\dagger} V\right) (1 - G_0 V)^{-1}$$

DMB at last

- S-matrix operator defined off-shell:

$$\hat{S}(E) = \left(1 - G_0(E)^\dagger V\right)\left(1 - G_0(E)V\right)^{-1}.$$

- Key identity (short derivation):

$$\frac{d}{dE}\mathrm{Tr}\log\hat{S} = \mathrm{Tr}\left[(G - G^\dagger) - (G_0 - G_0^\dagger)\right]$$

- But these are densities of state

$$\sum_i \frac{1}{E - E_i + i\epsilon} - \frac{1}{E - E_i - i\epsilon} = -2\pi i \sum_i \delta(E - E_i).$$

- This is DMB!

A simpler formula

- An alternate expression for the scattering operator

$$\hat{S} = 1 + (G_0 - G_0^*)T, \qquad T = V + VG_0V + VG_0VG_0V + \dots$$

- Leading term in log expansion

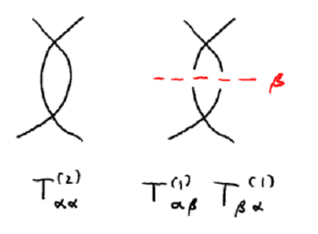
$$\delta Z = \int dE e^{-\beta E} \frac{1}{2\pi i} \frac{d}{dE} \mathrm{Tr} \log \hat{S} \approx \beta \int dE e^{-\beta E} \frac{1}{2\pi i} \mathrm{Tr}(G_0 - G_0^*)T$$

- Represent normalized trace over states by $\int d\alpha$

$$\delta Z = -\beta V \boxed{\int d\alpha e^{-\beta E_\alpha} \mathcal{M}_{\alpha\alpha}}$$

What did we throw away?

$$\log \hat{S} = (G_0 - G_0^*) \left[T - \frac{1}{2} T \, (G_0 - G_0^*) \, T + \dots \right]$$



- Higher terms in $\log S$ modify places where T goes on-shell
- At second order this is easy. Just use principal value prescription for $i\epsilon$ terms

$$V G_0 V - \frac{1}{2} V \, (G_0 - G_0^*) \, V = V \mathcal{P} G_0 V$$

Back to thermal field theory

- Start from Matsubara sum for one propagator:

$$\frac{1}{\beta}\sum_n \int \frac{dk}{2\pi}\,\frac{1}{\omega_n^2+E_k^2}, \quad E_k=\sqrt{k^2+m^2}$$

- After **Poisson resummation**:

$$\sum_l \int \frac{dk}{2\pi}\frac{d\omega}{2\pi}\,\frac{e^{-il\beta\omega}}{\omega^2+E_k^2}$$

- $l=0$: vacuum propagator.
- $l>0$ / $l<0$: poles at $\omega=\mp iE_k$.
- Summing: $\sum_{l\geq 1} e^{-l\beta E}=\frac{1}{e^{\beta E}-1}\equiv n_B(E)$.
- Each line = vacuum **"closed"** line + on-shell **"opened"** line

Opening free energy diagrams

- Pick r independent loop momenta. The rest is $\mathcal{M}$.

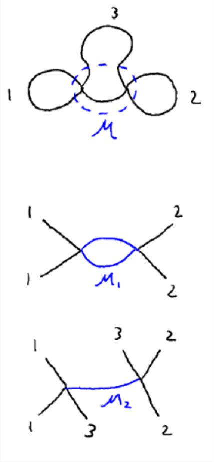
$$\delta f = \int \prod_{j=1}^{r} \frac{dk_j}{2\pi} \frac{d\omega_j}{2\pi} \frac{\sum_{l_j} e^{-il_j\beta\omega_j}}{\omega_j^2 + E_j^2} \mathcal{M}(\omega, k)\,.$$

- Choose which lines to open. Choice indexed by π
- Vacuum lines are lumped into on-shell $\mathcal{M}_\pi$

$$\delta f = \sum_{\pi} \int \prod_{j\in\pi} \frac{dk_j}{2\pi} \frac{n_B(E_j)}{2E_j} \mathcal{M}_\pi(\mp iE, k)\,.$$

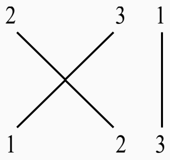
- This is just the simpler DMB formula!

$$\delta f = \boxed{\int d\alpha e^{-\beta E_\alpha} \mathcal{M}_{\alpha\alpha}}.$$



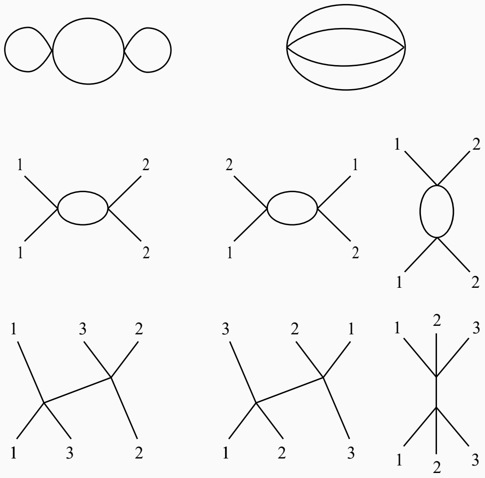
Cleaning up loose ends

- Expect $e^{-\beta E}$, but actually get products of **Bose factors** n_B
  - Comes from $n_B = \sum_n e^{-n\beta E}$.
  - Interpretable as extra "spectator" particles (connected in DMB sense).

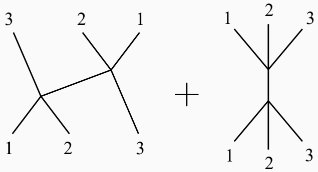


- $\mathcal{M}_\pi$ can have **poles/branch cuts**
  - Handled by higher terms in $\log S$.
  - Example: bubble diagram $\rightarrow$ double pole. Easy in thermal field theory

Examples in ϕ^4 theory



Calculating the melon



$$= 0?!$$

$$\frac{1}{-(E_1+E_2-E_3)^2+(k_1+k_2-k_3)^2+m^2}+(3\leftrightarrow 1)+(2\leftrightarrow 3) = -\frac{1}{-(E_1+E_2+E_3)^2+(k_1+k_2+k_3)^2+m^2}$$

- Related to 2-body factorization in sinh–Gordon.
- Does the melon really vanish?
- Using principal value prescription $\rightarrow$ survives only for collinear $k_1 = k_2 = k_3$.
- At the end of the day

$$f_{\text{melon}} = \# \int d\theta\, n_B(\theta)^2 + \# \int d\theta\, n_B(\theta)^3$$

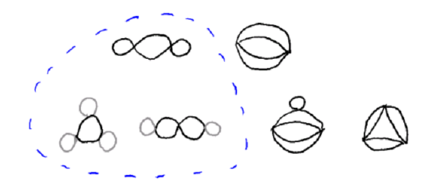
Back to effective strings

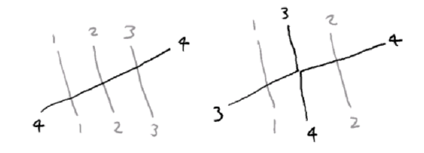
- Baratella–Miró–Gendy (2024): D=3 GGRT model
- Using DMB, matched known free energy to order β^{-6}

$$f = \frac{1}{l^2}\sqrt{1 - \frac{\pi l^2}{3\beta^2}} = \frac{1}{l^2} - \frac{\pi}{6\beta^2} - \frac{\pi^2 l^2}{72\beta^4} - \frac{\pi^3 l^4}{432\beta^6} - \dots$$

- In some (but not all) respects simpler than massive scalar
- Since all interactions are in terms of $\partial_L X \partial_R X$, only L and R scatter
- Melon diagram really does vanish

“Maximally singular” diagrams





- Caused a headache in early DMB, but simple in thermal field theory
- These are also the leading diagrams in vector-like large N

Matching to DMB

- The diagrams involve integrals $I_{,n} \equiv \int_\beta d^2p \frac{1}{(p^2+m^2)^{n+1}}$
- Can find exact formula

$$I_{,n} \propto \int \frac{dk}{2E} E^{-n} \sum_{l=0}^{n} \frac{(n+l)!}{(2E)^l (n-l)! l!} \left(-\frac{d}{dE}\right)^{n-l} n_B(E)$$

- Can we recover this in the DMB approach?
- Key result: For n singular old-fashioned propagators

$$\delta(E - H_0) \to \left(\frac{d}{dE}\right)^n \delta(E - H_0)$$

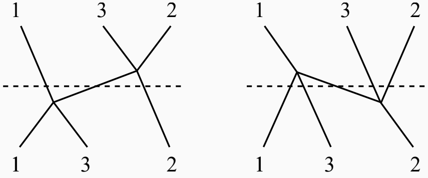
- Taking into account all time-orderings, full matching of formula

Example at second order (the bubble)

- How to get a derivative of a delta for the bubble diagram?
  - Recall old-fashioned propagator $G_0 = (E - H_0)^{-1}$
  - Recall the (corrected) second-order T matrix element is $V\mathcal{P}G_0V$
  - Multiply by an energy delta $\delta(E - H_0)$ at finite ϵ

$$(G_0 - G_0^*)\mathcal{P}G_0 = \frac{1}{2}\left(G_0^2 - G_0^{*2}\right) = -\frac{1}{2}\frac{d}{dE}(G_0 - G_0^*)$$

- Different 'time orderings' in OFPT have different number of singular propagators



Application to other models?

- So far, one particle per elementary field
- Models with bound states?
  - Lieb-Liniger with negative coupling
  - 2-body sector: Negative delta potential well
  - DMB is exactly solvable and accounts for bound state
- Does this apply in perturbative string theory?

What about other energy levels?

- In the integrable GGRT model, know $E_N(L)$ for all N
- What about the first excited state? Lüscher's single-particle formula

$$M(L) - m = -\frac{1}{2m}\int \frac{dp}{2\pi}\frac{1}{2E}e^{-\beta E}\mathcal{M}$$

- Verified in the $O(N)$ model. But what about 'cactus' diagrams?
- Lüscher's formula is like the simple version of DMB, can it be extended?
- Jeon–Ellis (1998) expand more general Σ in a DMB style

Application to kinetic theory?

- The Jeon-Ellis formulism can be understood in real time
- The thermal state is not special here, we may generalize n_B
- We can do "DMB" out of equilibrium
- Kadanoff-Baym generalizes the Boltzmann kinetic equation collision integral

$$\Sigma^{>} G^{<} - \Sigma^{<} G^{>}$$

- The lowest-order collision integral involves $|\mathcal{M}|^2$, can it be extended a la DMB?
- Carrington, Mrówczyński (2005), Jeon et al (2006)
- Application to wave turbulence?

Summing up

- DMB is an approach to calculating free energy and related quantities
- It can be derived from perturbative thermal field theory
- It can simplify or organize calculation in thermal field theory (e.g. the melon)
- It has been compared with a number of exact results (sinh-Gordon, GGRT, $O(N)$ model, Lieb-Liniger)
- It has an extension to self-energy that extends the Lüscher formula
- Speculatively: It may extend the Boltzmann collision integral

Thank you!

- Daniel Schubring, "Free energy from forward scattering in 1+1d," (2024), 2408.00729
- Baratella, Elias Miró, Gendy, "Thermodynamics from the S-matrix reloaded," (2024), 2408.06729
- S. Jeon and P. J. Ellis, "Multiple scattering expansion of the self-energy at finite temperature," (1998), hep-ph/9802246.
- R. Dashen, S. Ma, and H. J. Bernstein, "S-matrix formulation of statistical mechanics," Phys. Rev. 187 (1969) 345.